

CYBERSECURITY AWARENESS WORKSHEET - ANSWERS

Spot the Secure Website: HTTP vs HTTPS

This activity helps participants identify secure vs insecure websites by analysing URL structure, browser indicators, and trust signals.

The goal is to improve awareness of how to recognise potentially unsafe websites and understand that HTTPS alone does not guarantee legitimacy.

Section 1: Website Security Evaluation

Website	URL	Indicators	Secure? (Yes/No)
A	http://online-banking-login.example.com	No padlock, HTTP	No
B	https://university.edu/login	Padlock, HTTPS	Yes
C	https://secure-paypal-login.example.net	HTTPS but suspicious domain	Suspicious

Questions:

Which website is most secure? Why?

A → Uses HTTP (no encryption), no padlock, vulnerable to interception

B → Uses HTTPS, padlock icon, trusted domain (.edu)

C → HTTPS present but domain is misleading (possible phishing)

Why is HTTP considered unsafe?

HTTP → No encryption and is therefore vulnerable to interception

Does HTTPS always mean a site is trustworthy? Explain.

HTTPS → Uses encryption, but doesn't guarantee website safety/honesty/legitimacy

Section 2: Safe vs Unsafe Links

Evaluate the following links-

Task:

- Mark each link as Safe / Suspicious
- Identify one red flag for each suspicious link

1. <https://paypal.com/login>

Safe – Official domain

2. <https://paypal-secure-login.verify-now.example.com>

Suspicious – Fake domain structure (subdomain trick)

3. <https://uwa.edu.au>

Safe – trusted domain

4. <http://free-prizes.example.net>

Suspicious – HTTP + unrealistic content

Section 3: File Safety (Malware Awareness)

Determine whether these files are safe:

- invoice.pdf.exe
- report.docx
- update_patch_v2.zip
- photo.png

Questions:

- Which file is most suspicious? Why?
- What is a common trick used in malicious file names?

Invoice.pdf.exe – Suspicious (double extension hides executable file)

Report.docx – Safe (standard document format)

Update_patch_v2.zip – Suspicious (compressed file may contain malware)

Photo.png – Safe (standard image file)

Section 4: Password Strength Analysis

Rank the following passwords from weakest → strongest:

- | | |
|---|--------------|
| 1 | 123456 |
| 3 | password2024 |
| 2 | sam12345 |
| 4 | UWA!secure9 |
| 5 | Xy9#kL2pQ |

Questions:

- What makes a password strong?
- Why is password reuse dangerous?

Strong passwords → Length, randomness, symbols, numbers, mixed case

Password reuse → Increases risk if one account is compromised

Section 5: Social Engineering Scenarios

Scenario 1:

You receive a message from “IT Support” asking for your password to fix an issue.

Scenario 2:

A friend messages you urgently asking for money.

Questions:

- What are the warning signs?
- What would you do in each situation?

Scenario 1 →

Warnings signs – Asking for password directly / authority pressure

Correct action – Do not share password / contact official IT support

Scenario 2 →

Warning signs – Urgent request / emotional manipulation

Correct action – Verift identity through another method / don't send money

Immediately

Section 6: Quick Knowledge Check (Mixed Questions)

Multiple Choice:

1. What does HTTPS provide?
 - ☒ Encryption
 - ☐ Faster internet speed
 - ☐ Virus protection
2. What is phishing?
 - ☐ A type of malware
 - ☒ A social engineering attack
 - ☐ A firewall

True / False:

- HTTPS guarantees a website is completely safe (T / F) → **False**
- Strong passwords should include symbols and numbers (T / F) → **True**

Section 4: Reflection

Reflection section tailored to individual learning experience and acquired skills/knowledge

What new cybersecurity concept did you learn?

Which section did you find most difficult?

How will this activity change your online behaviour?
